Client
View Books | Register | Borrow | Return

Library Application
(API Gateway / Frontend)

Sidecar Proxy Layer
Service Discovery | mTLS | Routing | Health Checks | Retries | Logging

aws **AWS Cloud**

Book Service
(REST API)

Sidecar Proxy
(Routing | mTLS | Health)

Database
(RDS / DynamoDB)

Borrow Service
(REST API)

Sidecar Proxy
(Routing | mTLS | Health)

Database
(RDS / DynamoDB)

Redis Cache
- Book data
- Member data
- Cache hits / misses
- TTL expiration

mTLS
(Secure Communication)

GCP Cloud

Member Service
(REST API)

Sidecar Proxy
(Routing | mTLS | Health)

Database
(Cloud SQL / Firestore)

Borrow Service
(REST API)

Sidecar Proxy
(Routing | mTLS | Health)

Database
(Cloud SQL / Firestore)

⚙ Control Plane
Service Registry | Health Status | Traffic Weights | Configuration

Key Components

- Client
- Library Application
- Sidecar Proxy
- Library Services
- Database
- Redis Cache
- Control Plane
- AWS Cloud
- GCP Cloud

Data Flow

- Application / Data Plane
- Control Plane Traffic
- mTLS Communication
- Cache Flow (Redis)